

Lab Assignment 2 – Button LED Blink

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```
1  const int buttonPin = 2; // Push button pin
2  const int ledPin = 13;   // Built-in LED pin
3
4  void setup() {
5      pinMode(ledPin, OUTPUT);
6      // Fixed typo: Added the underscore for INPUT_PULLUP
7      pinMode(buttonPin, INPUT_PULLUP);
8
9      // Initialize Serial Monitor at 9600 baud rate
10     Serial.begin(9600);
11     Serial.println("System Ready. Press the button...");
12 }
13
14 void loop() {
15     // Button is pressed when the pin reads LOW (due to INPUT_PULLUP)
16     if (digitalRead(buttonPin) == LOW) {
17         digitalWrite(ledPin, HIGH); // Fixed typo: changed 'dagatal Wave' to 'digitalWrite'
18         Serial.println("Button Pressed -> LED ON");
19     } else {
20         digitalWrite(ledPin, LOW);
21         Serial.println("Button Released -> LED OFF");
22     }
23
24     delay(100); // Small delay to avoid flooding the Serial Monitor
25 }
```

[illegible]